



Predicting Used Car Prices

USING REGRESSION AND MACHINE LEARNING

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Executive Summary

This project proposes a pricing model that can effectively predict the price of used cars and can help Cars4U in devising profitable strategies using differential pricing. Predicting prices for used cars is significantly more complex than new cars, due to the many inconsistent factors that determine the price. Our results determined that the Tuned Decision Tree model to be most accurate, correctly predicting the price of a car almost 90% of the time. In the upcoming years, this model has risk of becoming outdated due to limited data on increasingly popular car features: electric power and automatic transmission. Additional data on such vehicles will need to be run through our model in order to remain efficient.

Problem Summary

The used vehicle market in India has a significant demand today, and has recently exceeded the new car market, based on a 2018-2019 study. There are even some car owners who are replacing their cars with used cars instead of buying a new one. New cars are priced fairly straightforward, as they primarily have standard parts, similar mileage, and manufacturer/dealership warranties. Due to the increasing demand for used cars, it is imperative to have a model to predict pricing for each unit, to prevent market instability and customers over or under paying for used cars. We need to implore various steps, such as establishing the importance of variables that determine a car's price, see if we can drop any insignificant variables, perform an Exploratory Data Analysis (EDA), and clean up the data. Then the key objective is to come up with a pricing model that can effectively predict the price of used cars to help businesses devise profitable strategies using differential pricing.

DATA EXPLORATION

We began the process of solving the problem by performing an EDA on our dataset to preprocess our data for the models. As the dataset consisted of both categorical and numerical variables, and our first step was to analyze the summary statistics of each. This provided insights on the 7,253 different cars such as the average year the car was built was in 2013, and the average kilometers driven was ~59,000. The average power of the vehicle was ~113BHP, but this was likely influenced by outliers as the 50th and 75th percentiles were only 94BHP and 138BHP, respectively. The most common type of transmission was manual, which accounted for approximately 71% of our dataset.

Next, we conducted both univariate and bivariate analyses on both the categorical and numerical values. During the univariate analysis, we found many outliers skewing our dataset; therefore, we applied log transformations to variables Kilometers_Driven and Price to help normalize their distributions, and added the transformed variables into the dataset. Our bivariate analysis provided insight on correlation between variables, as shown in the heat map below in Figure 1. Some of the strongest positive correlations were between Engine and Power, which makes sense as bigger engines allow for cars to have greater power. We also identified our transformed price variable, 'price_log', to have strong positive correlation with Engine, Power, and New_Price. Fuel efficiency, number of seats, and odometer readings proved to have the weakest correlation with the price of a car.

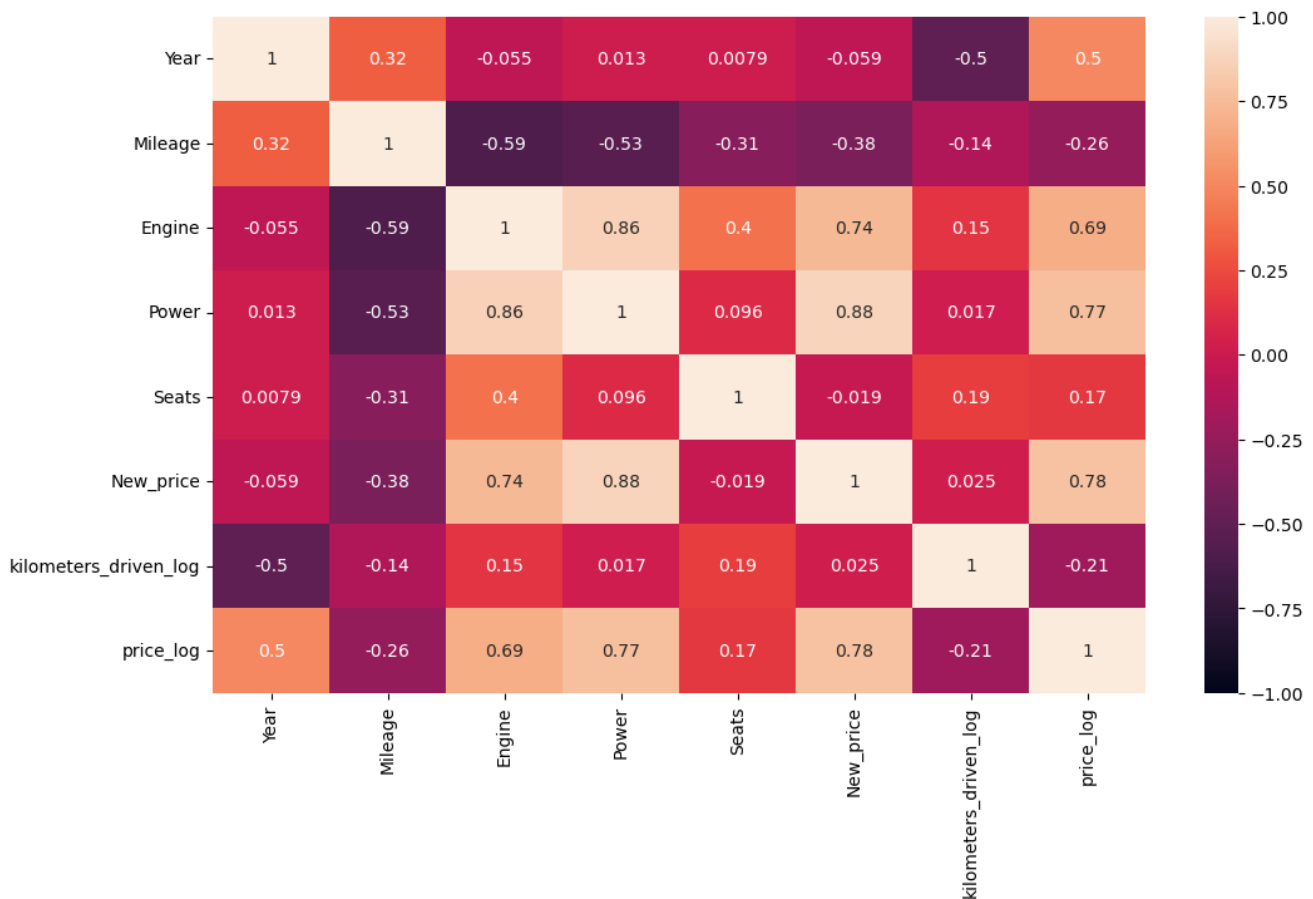


Figure 1

Our last step in EDA/preprocessing was to extract information from vague data and treat the missing values in the dataset. The variable Name contained 2041 unique values, which was over 25% of our total dataset. We extracted the brand name and model name separately to better prepare the data for our models. For missing value treatment, we imputed the missing data with median values of those variables.

Solution Summary

We started the process of building our models by separating independent and dependent variables, encoding categorical features, and splitting the data into train and test. We ran a series of regression algorithms, such as linear, ridge, lasso, decision tree, random forest, KNN as well as hyper-tuned decision tree and random forest. For each of these algorithms, we created a regression model, which we then fit to our train data, and obtained the scores of each. We also ranked the importance of features to these models, and consistently found the variables Power and Year to be most significant to the price of a car.

In order to improve our models, we can look at adding additional data such as number of accidents the vehicle has encountered, average time between engine service, average kilometers between oil change, age of tires, etc.

MEASURE OF SUCCESS

Upon evaluating our models, the Tuned Random Forest algorithm yielded the best results, with 89% accuracy on the test data and no signs of overfitting. This would be a valid solution for the problem due to its performance. Refer to Figure 2 below for more detailed analysis of our results for all models:

	Model	Train_r2	Test_r2	Train_RMSE	Test_RMSE
0	Linear Regression	0.862019	0.841759	4.200116	4.308038
1	Decision Tree	0.999997	0.806482	0.020887	4.764095
2	Ridge	0.859099	0.840006	4.244327	4.331835
3	Lasso	-1.021078	-0.247977	16.074723	12.098274
4	Random Forest	0.977502	0.897105	1.695990	3.473900
5	Tuned Decision Tree	0.924146	0.866172	3.114170	3.961805
6	Tuned Random Forest	0.921912	0.899379	3.159691	3.435291
7	KNN	0.851683	0.825385	4.354593	4.525438

Figure 2

Limitations & Recommendations for Implementation

One of the limitations for future performance of our model is lack of data on electric vehicles. Our dataset contained only two electric vehicles, and the electric vehicle market share is expected to rise above 20% by 2030. Another limitation is the amount of manual vs automatic transmissions in our dataset. The expected global market share of manual transmission cars by 2025 is 36%, but our data contained nearly 72% manual transmission

cars. As the market shares for both electric vehicles and automatic transmissions continue to grow, it is recommended to obtain more data on such vehicles for our models.

It is recommended that Tuned Random Forest model is implemented due to its very nearly equal train and test RMSE values, indicating the best fit out of all the models with 87% accuracy. The model performance is good enough for production, but can be improved by adding more numerical attributes, such as number of accidents, frequency of engine service, etc.

As the popularity among certain car features continues to change, it is recommended to rerun the algorithms with updated data based on latest market share for each of the categorical and numerical variables on an annual basis.